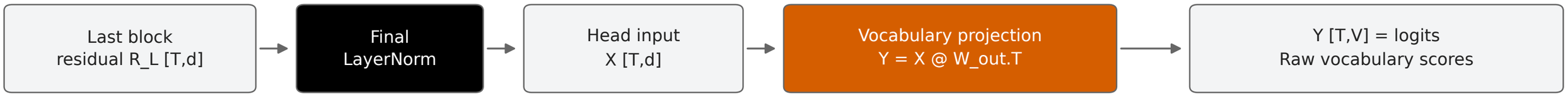
Vocabulary head: the projection already computes the scores

Existing B=1, T=2,048 forward; V=50,304; d=128 / 256 / 512. Conceptual diagram, not a new benchmark.

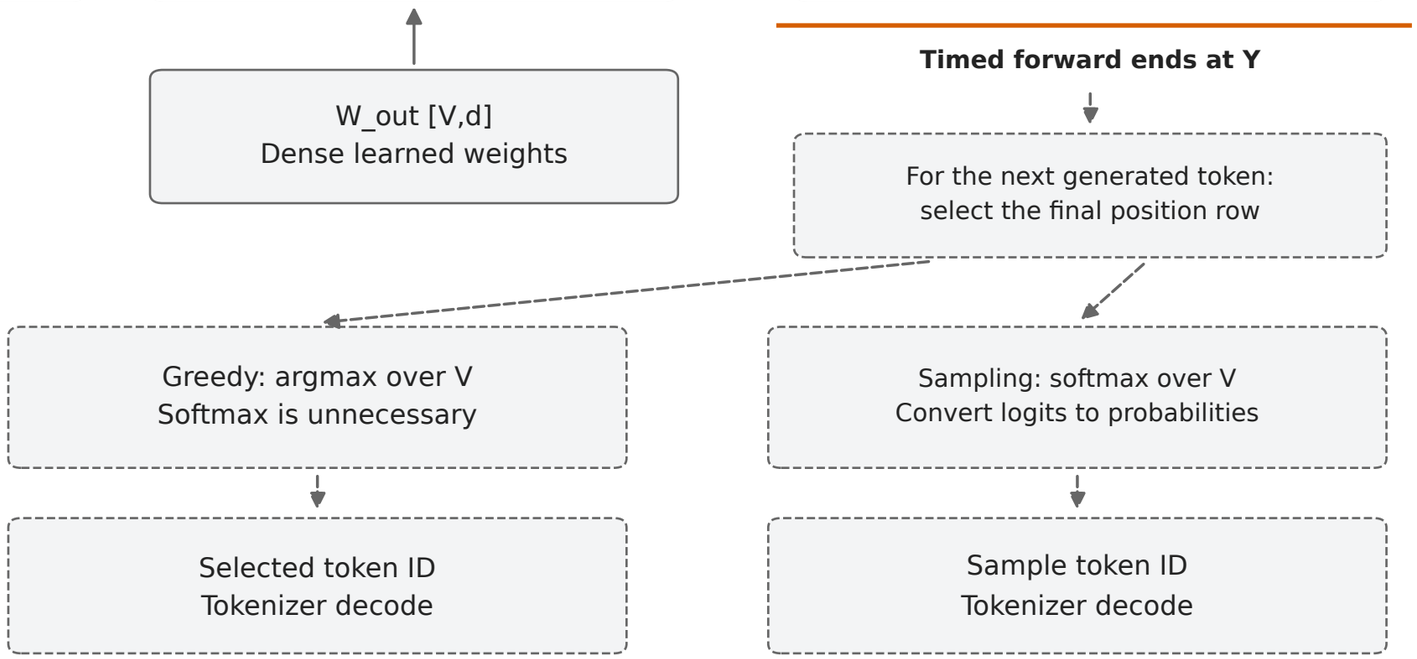
A. Current output path



Each score is one dot product:

$$Y_{t,v} = \sum_{j=1}^{d} X_{t,j} W_{v,j}$$

There is no additional Y -> scores dot product.



Other consumer: validation loss

Position logits + shifted token labels
feed cross-entropy (log-softmax + loss).
Also outside the timed forward.

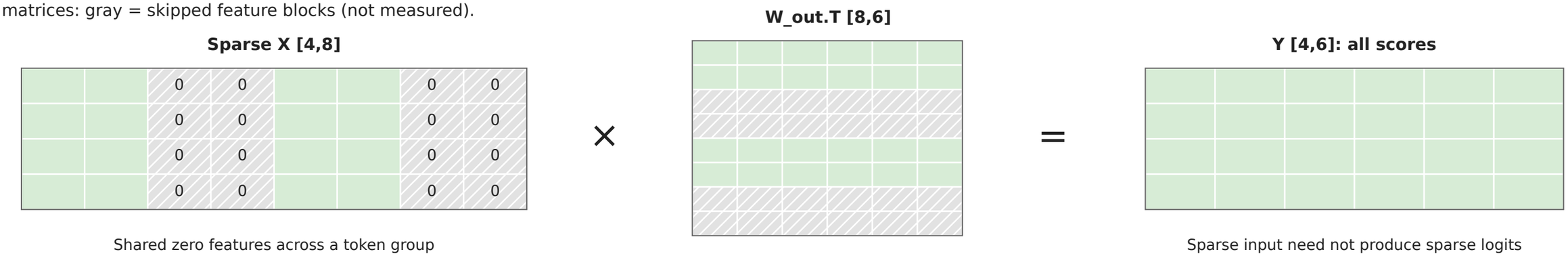
Solid = current measured path.
Dashed = downstream use or hypothetical intervention.
Head/norm colors match Figure 08.

B. Hypothetical sparse input: reduce features K, keep vocabulary N

Existing h/z gates do not gate X. A head-input gate would act after LayerNorm and require quality evaluation.



Toy matrices: gray = skipped feature blocks (not measured).



Shared zero features across a token group

Sparse input need not produce sparse logits

No gate/head changes were implemented. Smaller logical work does not establish speedup; full logits still require output storage.